

AUTOMATION SOFTWARE Moving Towards Minimising Human Involvement



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Testing is a very important part of the development life cycle, regardless of software or hardware. Every piece of equipment that leaves a manufacturing facility or each software that goes live is checked and rechecked for bugs. However, there are still some problems at times. Ever wondered why?

Testing is done in order to check if use cases work well with the equipment. However, in the case of manual testing, a person manually enters the inputs into the system and checks the output. Due to human involvement, exhaustive testing is not possible. More thorough testing could be done by automating the process. Create a test bench, code it for inputs, plug it in and you have a machine testing out the equipment for all possible use cases. Might not be as easy as it sounds, but it certainly is quicker.

Rohit Joshi, co-founder, Logic Fruit Technologies, explains some of the steps involved with custom-built solutions for testing. "We start with probing, where we connect to the device under test. Then, we acquire data from it. Data is stored and finally analysed. Today's high-speed devices require on-the-fly analysis, with near-zero latency. We achieve this by using field programmable gate arrays with speeds up to 16Gbps."

Importance of automated testing

High testing complexity with continuous daily tests of multiple application versions is just the beginning of the long list of use

cases for automated testing. Regression cycle and test cases with a short test window of 40 hours for end-to-end testing further adds to it. If you had to manually enter inputs, for 40 hours on an end, you would probably have a lateral head of hair.

Additionally, manag-

ing a team of testers spread across multiple environments including mobile devices, desktops, browsers and gaming consoles becomes a hassle when everybody is manually putting in numbers. HCL's electronics device automation testing (eDAT) services deployment showed that customers achieved 90 per cent reduction in test cycle time, per regression cycle. New devices can be added with minimal or negligible effort to a common framework of devices. It also reduces dependencies on human skill sets.

Taking data transfer bus as an example, automation platforms make it simple to test multiple devices at a time. The same user interface applies to different data buses like high-definition multimedia interface, peripheral component interconnect express or mobile industry processor interface physical layer specifications, which directly translates to productivity gains.

Automation is the trend

In testing any equipment, it has to be connected to some other equipment that can be automated. This automation is turning to more of a graphical interface for ease of access to you. This is reflected in other T&M equipment as well. "Test automation is maturing towards graphical user interface based test procedure implementation," says Leela Krishna Achutanna, senior manager - market development, aerospace and development, Rohde and Schwarz.

Asish Jain, application engineer, Keysight Technologies India Pvt Ltd, adds, "Now, non-programmers can also construct and configure test plans as per test requirements."

A majority of measurement equipment, be it an oscilloscope or a multimeter, are moving towards integration and control via a computer or smartphone. The move to control equipment via computers has resulted in the involvement of automation software to access equipment as well. Quickstep from R&S addresses the test au-

Some interesting features

Today's test automation software platforms allow selection of tests from an intuitive tree structure with multiple levels of detail. A sequencer also allows you to select the tests that you want to run. These results are also available in HTML or MS Excel for easy readability. Some of the other regulars would be real-time parameter changes in amplitude levels with options for single- and multi-lane device testing.